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Paradise Lost? Using Synthetic Historical Controls to Quantify Hawaii's Mandatory Quarantine on Tourism

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Summary This paper uses a novel causal time series method to estimate the causal impact of Hawaii's quarantine policy to combat the spread of COVID-19 on Hawaii's tourism industry.

Keywords: *Synthetic Controls, Policy Evaluation, COVID-19*

1. INTRODUCTION

Unlike other regions where economic contraction from COVID-19 emerged organically, through fear, voluntary distancing, or inconsistent mandates, Hawaii's economic/tourism collapse was the direct and intended consequence of state policy. On March 26 2020, the government took the extraordinary step of effectively sealing the state's borders, mandating a 14-day quarantine for all arrivals regardless of origin. In doing so, Hawaii enacted one of the few truly conscious economic shutdowns during the pandemic: a policy designed not just to mitigate risk, but to actively suppress travel and commerce. Ivanova (2020) documents that a third of Hawaii's labor force was unemployed within a month following the border closure, suggesting immediate economic impact.

Estimating the causal effect of this intervention presents a fundamental challenge: conventional comparative case study methods break down when all units are treated. Both difference-in-differences (DiD) and synthetic control methods (SCM) depend on the idea that one can compare a treated unit to a set of untreated units, under the assumption that, absent treatment, their outcomes would have followed either parallel trends or would be a convex combination of control units. In the COVID-19 context, this assumption is problematic. All U.S. states experienced pandemic-induced economic shocks, even if their formal policies varied. Thus, no state can be considered a credible untreated control for Hawaii. The same issue arises when extending to other island jurisdictions — every region was affected in some way. As Callaway and Li (2023), Bilgel (2022), and others argue, this contamination invalidates standard identifying assumptions in both DiD and SCM.

Despite this, researchers have applied SCM to study the effects of non-pharmaceutical interventions (NPIs) on health and economic outcomes during the pandemic [e.g., Yang (2021); Friedson et al. (2021); Gibson and Sun (2023); Breidenbach and Mitze (2021); Bulle et al. (2022)]. Others use regression based panel data approaches Ke and Hsiao (2022, 2023). These papers contribute valuable insights, but they all operate within the comparative case study framework where geographic control units exist, where certain cities or counties/regions would mandate masks and others would not. To overcome the necessity of an untreated donor pool, I use the *Synthetic Historical Control* (SHC) method, following the framework proposed by Chen et al. (2024). This method does not rely on geographical controls, instead relying only on the treated unit's history.

This paper makes both substantive and methodological contributions. Substantively, it contributes to the growing literature on the broader economic consequences of lockdown-style policies, beyond their direct effects on infection and mortality rates. Even if another pandemic does not occur soon, understanding the economic tradeoffs of extreme containment measures—such as mandatory quarantines and border closures—can help inform future policy in tourism-dependent economies. Moreover, given that most existing work has focused on interventions in mainland states [e.g., Friedson et al. (2021); Morgan et al. (2025); Gius (2024); Cohn et al. (2022); Dave et al. (2021, 2022)], the Hawaiian case offers a unique and underexplored setting.

Methodologically, this is the first applied paper (aside from the original SHC proposal) to implement the SHC method in a real-world setting. In doing so, it demonstrates the feasibility and value of SHC as a tool for policy evaluation when long time series are available but suitable donor units are unavailable or compromised.

The remainder of the paper is structured as follows. Section 2 introduces the empirical methodology. Section 3 describes the data. Section 4 presents the main findings. Section 5 concludes and discusses limitations.

2. METHODOLOGY

We know what happened given that Hawaii mandated two-week quarantines. We do not know what would have happened had they not done this. Given its novelty and radical deviation from SCM, this section summarizes the SHC method by Chen et al. (2024). For a comprehensive treatment of the method, including the theoretical proofs and asymptotic properties, readers are referred to Chen et al. (2024).

Notation

Let $\mathbb{R}$ denote the set of real numbers, and let $\|\cdot\|_1$ denote the usual ℓ_1 vector norm. Throughout, I denote sets using calligraphic letters (e.g., $\mathcal{T}, \mathcal{S}, \mathcal{N}$) for clarity. Scalars are represented using plain lowercase letters (e.g., h, t, n, m), and matrices are denoted by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{W}, \mathbf{Y}$). Given a vector $\mathbf{x} = (x_1, x_2, \ldots, x_T)^\top \in \mathbb{R}^T$ and an index set $\mathcal{I} \subseteq \{1, \ldots, T\}$, I write $\mathbf{x}_\mathcal{I} := (x_t)_{t\in\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$ to denote the subvector of $\mathbf{x}$ corresponding to indices in $\mathcal{I}$, preserving their original order.

Let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length, i.e., the number of periods used to construct the SHC match. Define the evaluation period as the final m months of the pre-treatment period:

$$\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1.$$

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. To reconstruct the evaluation window, we compare it against earlier segments of the treated unit's own pre-treatment trajectory.

Define the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$ as a collection of N overlapping length-m subvectors extracted from the smoothed pre-treatment series. Each column is a contiguous subvector of the form $\widetilde{\mathbf{y}}_{[i,i+m-1]}$, with i ranging over eligible start points in $\mathcal{T}_1$ that precede the evaluation window. The precise construction is given in Section~**??**.

2.1. Synthetic Historical Control Method

Potential Outcomes Following the Rubin potential outcomes framework, let y_t^1 and y_t^0 denote the potential outcomes for the treated unit at time $t \in \mathcal{T}$, corresponding to the outcomes under treatment and no treatment, respectively. The observed outcome at time t is $y_t = d_t y_t^1 + (1 - d_t) y_t^0$, where $d_t = \mathbf{1}\{t > T_0\}$.

The fundamental problem of causal inference is that for $t > T_0$, we observe only the treated potential outcome $y_t^1 = y_t$, while the untreated potential outcome y_t^0 remains unobserved. Let the SHC estimator for y_t^0 be denoted $\widehat{y}_t^0 := y_t^{\text{SHC}}$ for all $t \in \mathcal{T}_2$. Define the observed post-treatment outcome vector as $\mathbf{y}_{\text{post}} := (y_t^1)_{t \in \mathcal{T}_2}$ and the estimated counterfactual vector as $\mathbf{y}_{\text{post}}^0 := (\widehat{y}_t^0)_{t \in \mathcal{T}_2}$. The out-of-sample treatment effect is

$$\alpha_t := y_t^1 - y_t^0, \quad \text{for } t \in \mathcal{T}_2, \tag{2.1}$$

with estimator $\widehat{\alpha}_t := y_t - \widehat{y}_t^0$. Stacking over $t \in \mathcal{T}_2$ yields:

$$\widehat{\boldsymbol{\alpha}} := \mathbf{y}_{\text{post}} - \mathbf{y}_{\text{post}}^0.$$

The average treatment effect on the treated (ATT) is then given by:

$$\widehat{\text{ATT}} := \frac{1}{n} \sum_{t=T_0+1}^{T_0+n} \left(y_t - \widehat{y}_t^0 \right) = \frac{1}{n} \mathbf{1}^\top \widehat{\boldsymbol{\alpha}}, \tag{2.2}$$

where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.

2.2. The Model

The SHC method introduced by Chen et al. (2024) constructs a counterfactual using the treated unit's own pre-treatment time series.[1] SHC requires a few assumptions:

[1]SCM estimates the counterfactual by solving:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\operatorname{argmin}} \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2, \quad \forall t \in \mathcal{T}_1, \tag{2.3}$$

Assumption 2.1. (Error Properties and Regressor Independence) *The error term satisfies* $\mathbb{E}[\varepsilon_t] = 0$ *and* $\mathbb{E}[\varepsilon_t^2] < \infty$ *for all* $t \in \mathcal{T}$.

Or the mean of the residuals is 0 in expectation. This fairly standard assumption says any unmodeled fluctuations or noise in the outcome data are purely random and do not systematically bias the estimation of the counterfactual.

Assumption 2.2. (Latent Component Smoothness and Matching Condition) *The latent trend* $\widetilde{y}(t)$ *is* $(H+1)$ *times differentiable for some integer* $H \geq 3$, *with derivative bounded as* $\left|\widetilde{y}^{(H+1)}(t)\right| < \varepsilon$ *uniformly over* $t \in \mathcal{T}_1$. *There exists a convex weight vector* $\mathbf{w}^* \in \Delta^{N-1}$ *such that the evaluation subvector* $\mathbf{y}_{eval}$ *can be exactly matched by a convex combination of donor segments:* $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*$.

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where $\mathbf{Y}_0 \in \mathbb{R}^{T_0 \times N_0}$ is the donor matrix of unexposed units and $\Delta^{N_0} = \left\{\mathbf{w} \in \mathbb{R}^{N_0}_{\geq 0} : \|\mathbf{w}\|_1 = 1\right\}$ is the probability simplex.

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